

On the mathematical foundations of the self-consistent clustering analysis for non-linear materials at small strains

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Highlights

- Stream-lined derivation of the clustered Lippmann–Schwinger equation.
- Reformulation of the clustered Lippmann–Schwinger equation as an Eyre–Milton equation.
- Well-posedness of discretized equations for small strain hardening materials.
- Proof of convergence of discrete solutions upon cluster refinement.
- Numerical investigations of the convergence rate.

Abstract

We investigate, both mathematically and numerically, the self-consistent clustering analysis recently introduced by Liu–Bessa–Liu and, independently, by Wulfinghoff–Cavaliere–Reese. We establish, in the small strain setting and non-softening material behavior, existence and uniqueness of the solution to the discretized equations for fixed (possibly anisotropic) reference material, by a constructive method. Furthermore, we establish convergence of the solution to the discretized equation to the continuous solution upon refinement of the clusters. Thus we generalize the work of Tang–Zhang–Liu to spatial dimensions larger than 1.

We explore the specific structure of the Lippmann–Schwinger equation governing clustering analysis, proving strict equivalence to a problem of Eyre–Milton type. For the latter formulation, existence and uniqueness are easily established based on recent progress in the understanding of polarization schemes in FFT-based computational homogenization methods.

Last but not least, for elasto-viscoplastic constituents we clarify the relationship of the self-consistent clustering analysis to the transformation field analysis of Dvorak–Benveniste. Our theoretical considerations are confirmed by pertinent numerical experiments.

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1. Introduction

Modeling approaches for materials with heterogeneous microstructure can be roughly classified into three types:

1. Phenomenological models,
2. Analytical models,
3. Order reduced models.

(Note however, that the classification is not strict.)

For the phenomenological models approach [1–3], a specific material model is postulated whose parameters are subsequently identified either by real or computational experiments based on minimizing an error function. This approach unites speed, flexibility, robustness and is certainly most used for industrial applications. A major restriction of phenomenological models is their dependence on the quality of the postulated material model — it can be challenging to design a material model that can account for all relevant mechanical effects induced by anisotropy, strain rate dependence etc. Notice, however, that the phenomenological models approach has gained new impetus by transferring insights from neural networks [4–8] by postulating a neural-network based model which is subsequently “trained”.

Analytical methods rely upon closed form solutions for specific heterogeneous microstructures [9–14] or estimates obtained from variational principles [15–20], taking into account the material behavior of the constituents and “statistical” properties of the microstructure (which should be known in advance) and the fields (mean and variance). Compared to numerical approaches, analytical methods are typically very fast, but their accuracy can deteriorate for highly-contrasted, nonlinear materials or in case of localization. Also, for heterogeneous materials whose morphology strongly differs from matrix-inclusion microstructures, analytical methods may have trouble, see Schneider et al. [21] for an example.

The mathematical theory of homogenization [22] has triggered a multitude of computational homogenization approaches [23]. In this context, mechanical problems are solved on heterogeneous material volumes which are considered representative for the microstructure. In concurrent multiscale simulations [24], the material behavior of each macroscopic integration point is computed by solving an associated micro-scale problem on a corresponding unit cell. To alleviate the often tremendous computational burden of concurrent multiscale simulations, model order reduction techniques were developed, where the full-field unit cell simulations are replaced by a numerically informed surrogate model.

The first such technique, the transformation field analysis (TFA), was introduced by Dvorak–Benveniste [25] and subsequently refined [26,27]. TFA applies to small strain materials of elasto-viscoplastic type. The (visco)plastic strains are approximated as piecewise-constant on a pre-chosen partition of the cell, and a corresponding database is generated by solving dedicated elastic eigenstrain problems. This database is essential for the reduced TFA model, which is reminiscent of crystal (visco)plasticity models — the plastic strains on the partitions extend the slips in classical crystal plasticity models. By construction, the TFA captures the elastic behavior of the composite correctly. However, in the nonlinear regime, the stress predicted by the TFA could be much larger than the “full-field” stress. Michel–Suquet [28] attributed this behavior to the piece-wise uniformity of the plastic strain, and introduced a non-uniform transformation field analysis (NTFA), where the plastic strain fields could be arbitrary. In this way, the approximation error of the plastic strain could be made as small as necessary. However, in contrast to the original TFA, for evaluating the material laws either the fully resolved fields would have to be accessed, or suitably chosen “effective” material laws need to be postulated. (Only the case of small strain viscoelasticity [29,30] can be treated explicitly.) Different ideas on combining low approximation error and computational speed have been proposed, for instance mixed variational principles [31,32] and series expansions [33–35].

The self-consistent clustering analysis (SCA) was introduced by Liu–Bessa–Liu [36]. They introduced (an incremental version of) the clustered Lippmann–Schwinger equation in an ad-hoc manner for small strain inelastic materials. They propose k -means clustering of the strain concentration tensor for cluster identification, and to choose the reference material in a self-consistent manner. The method was subsequently applied to crystal viscoplasticity at small [37] and finite strains [38], ductile materials [39] and the data-driven analysis of materials under uncertainty [40]. Liu–Fleming–Liu [41] presented an extension to small strain elastic–plastic materials coupled to a non-local damage model. Tang–Zhang–Liu [42] proved the convergence of the SCA approximation in one spatial dimension, and presented numerical convergence studies in one and two dimensions. Furthermore, they introduced a variant of the SCA, where the unit cell is embedded into a larger reference medium.

Wulfinghoff–Cavaliere–Reese [43] introduced an extension of the variational principle of Hashin–Shtrikman by model order reduction ideas which turned out to be equivalent to the SCA. However, their approach differs in various aspects. Firstly, their treatment is based on the Hashin–Shtrikman variational principle. In particular, they emphasize the connection to classical analytical estimates of Talbot–Willis-type [19]. Secondly, their clustering is based on snapshots of the nonlinear mechanical behavior, together with a projection procedure onto the macroscopic strain. (In contrast, Liu–Bessa–Liu [36] cluster based on k -means clustering [44]). Last but not least, Wulfinghoff et al. chose the reference material with the help of the secant stiffness (and not based on a self-consistent estimate as Liu et al. [36]).

Contributions

In this work (Section 2), we give a streamlined derivation of the clustered Lippmann–Schwinger equation based on the Hashin–Shtrikman variational principle. In contrast to Wulfinghoff–Cavaliere–Reese [43], there is no need to condense variables a posteriori, because the strain tensor field serves as the primary unknown from the very beginning.

We present a mathematical proof for the existence and uniqueness of solutions to the clustered Lippmann–Schwinger equation for fixed reference material (but of possibly arbitrary anisotropy), arbitrary spatial dimension and small strain materials exhibiting strain-hardening behavior (see Eqs. (2.1) and (2.2) for the precise requirements). Furthermore, we establish convergence of the solutions of the clustered Lippmann–Schwinger equation to the solution of the continuous Lippmann–Schwinger equation upon cluster refinement. The details comprise Section 3.

The key technical idea for this work is recent progress in understanding polarization schemes popular in FFT-based computational micromechanics. More precisely, we show that the clustered Lippmann–Schwinger equation can be equivalently rewritten as a clustered Eyre–Milton equation (3.6), for which Banach’s fixed point Theorem can be applied to show existence and uniqueness of solutions. The contraction property of a particular nonlinear operator is also crucial for establishing convergence upon cluster refinement.

Thus, we extend the one-dimensional analysis of Tang–Zhang–Liu [42] to higher dimensions and the work of Brisard–Dormieux [45,46], who investigated the clustered Lippmann–Schwinger equation for identically shaped voxel clusters, to arbitrary cluster shape and nonlinear material behavior.

Last but not least, we establish the close connection of the SCA to the original TFA of Dvorak in the context of small-strain elasto-viscoplastic materials. Both SCA and TFA can be brought into a form where the two schemes are identical up to a single operator (of Eshelby–Green type): for SCA this operator is clustered, for TFA, it is non-clustered. Thus, we aid in putting the SCA into the proper perspective concerning modern model order reduction methods in computational micromechanics. The SCA serves in connecting analytical estimates and full-field simulations (containing both as limiting cases). Furthermore, SCA is similar in spirit to the TFA, inheriting the advantages, but relieving some restrictions for applicability concerning the class of material behavior which can be handled.

This work is concluded by numerical investigations (Section 5) on the convergence behavior of SCA for E-glass-reinforced polyamide microstructures.

2. On the Hashin–Shtrikman variational principle and its discretization by clustering

2.1. The corrector problem and its equivalent Lippmann–Schwinger/Hashin–Shtrikman formulations

Let Y be a rectangular cell in $\mathbb{R}^d$ and let a strain energy potential $\psi : Y \times \text{Sym}(d) \rightarrow \mathbb{R}$ be given, where $\text{Sym}(d)$ denotes the vector space of symmetric $d \times d$ -matrices. Suppose that ψ is measurable in the first variable, and continuously differentiable in the second variable. For any $\varepsilon \in \text{Sym}(d)$, the partial derivative $\frac{\partial \psi}{\partial \varepsilon}(x, \varepsilon)$ is called the stress at $x \in Y$ induced by the strain ε .

To put our framework in perspective, the strain energy potential we consider can either be thought of as a small-strain hyper-elastic material model or may arise from a time discretization of an inelastic small-strain material where the evolution of the internal variables is “frozen”.

For further progress, we assume further smoothness and coercivity properties of ψ . Suppose there are positive constants α_- and α_+ , s.t.

$$\left[\frac{\partial \psi}{\partial \varepsilon}(x, S) - \frac{\partial \psi}{\partial \varepsilon}(x, T) \right] : (S - T) \geq \alpha_- \|S - T\|^2 \quad (2.1)$$

and

$$\left\| \frac{\partial \psi}{\partial \varepsilon}(x, S) - \frac{\partial \psi}{\partial \varepsilon}(x, T) \right\| \leq \alpha_+ \|S - T\| \quad (2.2)$$

hold for (almost) any $x \in Y$ and arbitrary $S, T \in \text{Sym}(d)$. Here $\cdot$ denotes the Frobenian inner product

$$\text{Sym}(d) \times \text{Sym}(d) \ni (S, T) \mapsto S : T \equiv \text{tr}(ST),$$

and $\|\cdot\|$ stands for the induced norm on $\text{Sym}(d)$. (2.1) states that $\frac{\partial \psi}{\partial \varepsilon}$ is a uniformly strongly monotone operator, which is furthermore uniformly Lipschitz continuous. From a mechanical point of view, the first condition (2.1) states that stress increases with strain in a strongly monotone manner. (For twice differentiable strain energy potentials, this is equivalent to a positive, uniform lower bound on the eigenvalues of the material tangent.) The second assumption (2.2) is typically satisfied for small strain materials, because the initial linear elastic material response bounds the nonlinear material response from above.

Under this assumption (2.2), $\frac{\partial \psi}{\partial \varepsilon}$ gives rise to a well-defined (Nemytskii) operator

$$\sigma : L^2(Y; \text{Sym}(d)) \rightarrow L^2(Y; \text{Sym}(d)), \quad \varepsilon \mapsto \frac{\partial \psi}{\partial \varepsilon}(\cdot, \varepsilon(\cdot)), \quad (2.3)$$

which we call the stress operator. The stress operator encodes the material law. For given strain field ε , $\sigma(\varepsilon)$ is precisely the associated stress field.

In computational homogenization on periodic unit cells, for prescribed macroscopic strain $E \in \text{Sym}(d)$, a periodic displacement fluctuation

$$u \in H_{\#}^1(Y; \mathbb{R}^d) = \{u \in H^1(Y; \mathbb{R}^d) \mid u \text{ periodic, } n \cdot \nabla u \text{ anti-periodic on } \partial Y, \langle u \rangle_Y = 0\}, \quad (2.4)$$

where n denotes the outward-pointing unit normal on ∂Y and $\langle \cdot \rangle_Y$ refers to the (component-wise) mean value operator on Y , is sought, solving the corrector problem

$$\text{div } \sigma(E + \nabla^s u) = 0, \quad (2.5)$$

where $\nabla^s : H_{\#}^1(Y; \mathbb{R}^d) \rightarrow L^2(Y; \text{Sym}(d))$ denotes the symmetrized gradient and $\text{div} : L^2(Y; \text{Sym}(d)) \rightarrow H_{\#}^{-1}(Y; \mathbb{R}^d)$ is the negative of its adjoint operator. Under the assumptions (2.1) and (2.2), existence and uniqueness of the solution $u \in H_{\#}^1(Y; \mathbb{R}^d)$ to (2.5) can be established [47,48].

Emphasizing the explicit dependence of the solution to (2.5) on the macroscopic strain E by including a subscript u_E , the primary quantity of interest is the effective stress

$$\Sigma : \text{Sym}(d) \rightarrow \text{Sym}(d), \quad E \mapsto \langle \sigma(E + \nabla^s u_E) \rangle_Y,$$

which constitutes the effective material law, “homogenized” from σ .

Both for the development of estimates and numerical methods for effective material behavior it is useful to reformulate (2.5) in terms of a Lippmann–Schwinger equation. A reference material is a linear mapping $\mathbb{C}^0 \in L(\text{Sym}(d))$ which is symmetric w.r.t. the Frobenian inner product on $\text{Sym}(d)$ and furthermore positive definite, i.e.

$$S : \mathbb{C}^0 : S > 0 \quad \text{for any } S \in \text{Sym}(d), \quad S \neq 0.$$

Any such reference material induces an inner product on $L^2(Y; \text{Sym}(d))$ by

$$(S, T) \mapsto \langle S : \mathbb{C}^0 : T \rangle_Y \quad (2.6)$$

with associated norm $S \mapsto \|S\|_{\mathbb{C}^0}$. Different reference materials give rise to equivalent norms on $L^2(Y; \text{Sym}(d))$.

By the Korn–Poincaré inequality [49] the set $\nabla^s H_{\#}^1(Y; \mathbb{R}^d)$, which is the image of $H_{\#}^1(Y; \mathbb{R}^d)$ w.r.t. the symmetrized gradient, constitutes a closed subspace of $L^2(Y; \text{Sym}(d))$. $\Gamma^0 : \mathbb{C}^0$ is precisely the orthogonal projector [47] onto $\nabla^s H_{\#}^1(Y; \mathbb{R}^d)$ w.r.t. $\langle \cdot, \cdot \rangle_{\mathbb{C}^0}$, where the Eshelby–Green operator

$$\Gamma^0 \in L(L^2(Y; \text{Sym}(d))), \quad \Gamma^0 = \nabla^s (\text{div } \mathbb{C}^0 \nabla^s)^{-1} \text{div}$$

is defined via the inverse of the linear elastic partial differential operator

$$H_{\#}^1(Y; \mathbb{R}^d) \rightarrow H_{\#}^{-1}(Y; \mathbb{R}^d), \quad u \mapsto \text{div } [\mathbb{C}^0 : \nabla^s u]$$

with homogeneous coefficients defined by the reference material $\mathbb{C}^0$. Here $L(V)$ denotes the Banach space of bounded linear operators on any Banach space V , equipped with the induced operator norm.

For fixed reference material $\mathbb{C}^0$, the associated Lippmann–Schwinger equation reads

$$\varepsilon + \Gamma^0 : (\sigma(\varepsilon) - \mathbb{C}^0 : \varepsilon) = E, \quad \varepsilon \in L^2(Y; \text{Sym}(d)). \quad (2.7)$$

It can be shown [47] that under the assumptions (2.1) and (2.2), there exists a unique solution $\varepsilon \in L^2(Y; \text{Sym}(d))$ to the Lippmann–Schwinger equation (2.7). Furthermore, the solution is independent of the reference material $\mathbb{C}^0$, and can be written explicitly

$$\varepsilon = E + \nabla^s u_E$$

in terms of the displacement solution $u_E \in H_{\#}^1(Y; \mathbb{R}^d)$ to the corrector problem (2.5).

From the point of view of mean field methods and in terms of numerics, the basic advantage of the Lippmann–Schwinger equation (2.7) compared to the corrector equation (2.5) lies in its domain of definition. The Lippmann–Schwinger equation is defined on the L^2 scale. In particular, discontinuous fields can be used as inputs of the operators. In contrast, compatible displacements need to be constructed for use in conjunction with (2.5).

For estimates, a variational formulation of (2.7) is useful. Denote by $\alpha_{\pm}^0$ the largest and smallest eigenvalues of $\mathbb{C}^0$, respectively. If $\alpha_+^0 < \alpha_-$ for the constant α_- entering (2.1), the operator $\sigma - \mathbb{C}^0$ defines a strongly monotone (and Lipschitz continuous) operator on $L^2(Y; \text{Sym}(d))$. Thus, it can be inverted to arrive at the equation

$$(\sigma - \mathbb{C}^0)^{-1} \tau + \Gamma^0 : \tau = E, \quad (2.8)$$

equivalent to the Lippmann–Schwinger equation (2.7) via the correspondence $\tau = \sigma(\varepsilon) - \mathbb{C}^0 : \varepsilon$.

The function

$$\varphi : L^2(Y; \text{Sym}(d)) \rightarrow \mathbb{R}, \quad \varepsilon \mapsto \frac{1}{|Y|} \int_Y \psi(x, \varepsilon(x)) - \frac{1}{2} \varepsilon(x) : \mathbb{C}^0 : \varepsilon(x) dx$$

is continuously (Frechét) differentiable and serves as a potential for $\sigma - \mathbb{C}^0$, i.e.

$$\frac{1}{|Y|} \int_Y D\varphi[\varepsilon](x) : \delta\varepsilon(x) dx \equiv \left. \frac{d}{ds} \varphi(\varepsilon + s\delta\varepsilon) \right|_{s=0} = \frac{1}{|Y|} \int_Y [\sigma(\varepsilon)(x) - \mathbb{C}^0 : \varepsilon(x)] : \delta\varepsilon(x) dx$$

holds for all $\varepsilon, \delta\varepsilon \in L^2(Y; \text{Sym}(d))$. Put differently,

$$D\varphi[\varepsilon] = \sigma(\varepsilon) - \mathbb{C}^0 : \varepsilon \quad \text{holds for all } \varepsilon \in L^2(Y; \text{Sym}(d)),$$

which we write succinctly as $D\varphi = \sigma - \mathbb{C}^0$. Then, the Legendre–Fenchel dual [50]

$$\varphi^* : L^2(Y; \text{Sym}(d)) \rightarrow \mathbb{R}, \quad \tau \mapsto \sup_{\varepsilon \in L^2(Y; \text{Sym}(d))} \langle \tau : \varepsilon \rangle_Y - \varphi(\varepsilon)$$

is also continuously differentiable with derivative

$$D\varphi^* = (\sigma - \mathbb{C}^0)^{-1}.$$

As first observed by Willis [18], using φ^* Eq. (2.8) can be obtained as the Euler–Lagrange equation of the Hashin–Shtrikman functional

$$\text{HS} : L^2(Y; \text{Sym}(d)) \rightarrow \mathbb{R}, \quad \tau \mapsto \varphi^*(\tau) + \frac{1}{2} \langle \tau : \Gamma^0 : \tau \rangle_Y - \langle \tau \rangle_Y : E, \quad (2.9)$$

where the second summand involves a symmetric and non-negative bilinear form, which is apparent from the formula

$$\langle S : \Gamma^0 : T \rangle_Y = \langle \mathbb{D}^0 : S, \Gamma^0 : \mathbb{C}^0 : (\mathbb{D}^0 : T) \rangle_{\mathbb{C}^0}, \quad S, T \in L^2(Y; \text{Sym}(d)),$$

where $\mathbb{D}^0 = (\mathbb{C}^0)^{-1}$ denotes the reference compliance, and the orthogonal projector property of $\Gamma^0 : \mathbb{C}^0$ is used.

Our derivation of the Hashin–Shtrikman principle (2.9) was carried out for sufficiently small reference material $\mathbb{C}^0$ to ensure invertibility of $\sigma - \mathbb{C}^0$. Alternatively, $\mathbb{C}^0$ can be chosen sufficiently large, s.t. $\sigma - \mathbb{C}^0$ is “sufficiently negative”. More precisely, if the smallest eigenvalue of $\mathbb{C}^0$ satisfies $\alpha_-^0 > \alpha_+$, where α_+ is the Lipschitz constant of σ , the operator $\mathbb{C}^0 - \sigma$ is strongly monotone, and the considerations above can be carried out similarly, with the notable difference that the Hashin–Shtrikman variational principle (2.9) is no longer a minimization, but a maximization principle.

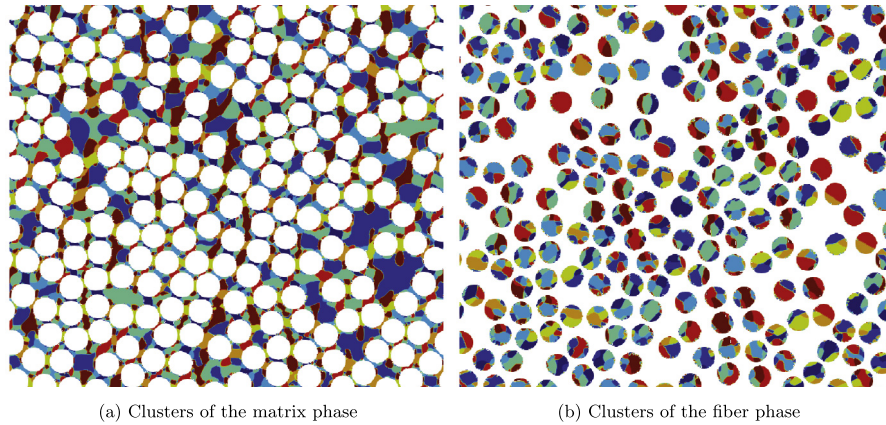


Fig. 1. A uni-directionally fiber reinforced composite microstructure with 50% fiber volume content, clustered into $8 + 8 = 16$ clusters. Each color corresponds to a cluster. The clustering was obtained by randomly initialized k -means clustering based on the strain localization tensor. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

For general reference material, the formal first order conditions of the Hashin–Shtrikman functional (2.9) are still equivalent to the Lippmann–Schwinger equation (2.7), but the Hashin–Shtrikman functional is unbounded from above and from below, in general.

In the continuous context, the last remark is not problematic, as the solution to the Lippmann–Schwinger equation (2.7), and, thus, the critical point of the Hashin–Shtrikman functional, does not depend on the chosen reference material $\mathbb{C}^0$.

2.2. Discretization of the Hashin–Shtrikman functional by clustering

Suppose a cluster decomposition of Y is given, i.e. a collection $\{\Omega_i\}_{i=1}^N$ of non-empty, open subsets of Y which are mutually disjoint and whose closure covers Y . For an example of a clustering, see Fig. 1. Notice that we do not assume the clusters to be connected or convex.

To each cluster Ω_i we associate its characteristic function χ_i , i.e.

$$\chi_i(x) = \begin{cases} 1, & x \in \Omega_i, \\ 0, & \text{otherwise.} \end{cases}$$

We further assume that the cluster decomposition $\{\chi_i\}$ is compatible to the given stress operator. Formally, we require that

$$\sigma(\chi_i \varepsilon) = \chi_i \sigma(\varepsilon) \quad (2.10)$$

holds almost everywhere and for all $i = 1, \dots, N$ and $\varepsilon \in V_N$. Specifically, (2.10) ensures that the cluster decomposition respects the heterogeneity of the stress operator. More precisely, if a multi-phase material is considered, consisting of K phases $Y_1, \dots, Y_K$, and σ is homogeneous on each Y_k , then (2.10) ensures that for each Ω_i there is a Y_k , s.t. $\Omega_i \subseteq Y_k$.

For the succeeding analysis, the slightly more general condition (2.10) is sufficient.

For fixed cluster decomposition $\{\Omega_i\}$ we associate the subspace

$$V_N = \{S \in L^2(Y; \text{Sym}(d)) \mid S \text{ is homogeneous on } \Omega_i \text{ for all } i = 1, \dots, N\} \quad (2.11)$$

of $L^2(Y; \text{Sym}(d))$. The linear operator

$$Q_N \in L(L^2(Y; \text{Sym}(d))), \quad S \mapsto \sum_{i=1}^N \frac{1}{\langle \chi_i \rangle_Y} \chi_i \langle S \chi_i \rangle_Y \quad (2.12)$$

defines a projector onto V_N , which is furthermore orthogonal for the inner product $\langle \cdot, \cdot \rangle_{\mathbb{C}^0}$, independently of the chosen reference material $\mathbb{C}^0$. Furthermore, by the compatibility of the stress operator σ and the cluster

decomposition $\{\Omega_i\}$, (2.10) is equivalent to the commutativity of Q_N and σ on V_N , i.e.

$$Q_N \sigma(\varepsilon) = \sigma(Q_N \varepsilon) \quad \text{for all } \varepsilon \in V_N. \quad (2.13)$$

Liu–Bessa–Liu [36] proposed to investigate a Galerkin discretization based upon a fixed cluster decomposition $\{\Omega_i\}$ which can be derived, cf. Wulfinghoff–Cavaliere–Reese [43], from the Hashin–Shtrikman variational principle for fixed reference material $\mathbb{C}^0$

$$HS(\tau) \equiv \varphi^*(\tau) + \frac{1}{2} \langle \tau : \Gamma^0 : \tau \rangle_Y - \langle \tau \rangle_Y : E \longrightarrow \min_{\tau \in V_N}, \quad (2.14)$$

where we fix $\mathbb{C}^0$ to be sufficiently small as specified in the previous section. The corresponding Euler–Lagrange equations read

$$Q_N : (\sigma - \mathbb{C}^0)^{-1} Q_N : \tau_N + Q_N : \Gamma^0 : Q_N : \tau_N = E, \quad \tau_N \in V_N.$$

With the help of the compatibility condition (2.10), equivalently we seek $\tau_N \in V_N$ satisfying the equation

$$(\sigma - \mathbb{C}^0)^{-1} \tau_N + \Gamma_N^0 : \tau_N = E \quad (2.15)$$

for $\Gamma_N^0 = Q_N \Gamma^0 Q_N$, which differs from the original Hashin–Shtrikman equation (2.8) only in terms of the Eshelby–Green operator. As $\sigma - \mathbb{C}^0$ is strongly monotone, equivalently, we can also seek $\varepsilon_N \in V_N$ solving the clustered Lippmann–Schwinger equation

$$\varepsilon_N + \Gamma_N^0 : (\sigma(\varepsilon_N) - \mathbb{C}^0 : \varepsilon_N) = E, \quad (2.16)$$

where ε_N and τ_N are related by $\tau_N = \sigma(\varepsilon_N) - \mathbb{C}^0 : \varepsilon_N$. The numerical approaches of Liu–Bessa–Liu [36] and Wulfinghoff–Cavaliere–Reese [43], rest upon two observations:

1. The clustered Lippmann–Schwinger equation (2.16) is well-defined for any reference material $\mathbb{C}^0$, in particular for cases where the derivation by the Hashin–Shtrikman variational principle can no longer be mathematically justified.
2. If σ is homogeneous on the cluster decomposition $\{\Omega_i\}$, the clustered Lippmann–Schwinger equation can be understood as an equation on V_N . In particular, for three spatial dimensions, the clustered Eshelby–Green operator Γ_N^0 can be represented by a $6N \times 6N$ matrix, whose entries can be precomputed numerically (depending only on the clustering and not on the material law).

Concerning the solutions to the clustered Lippmann–Schwinger equation (2.16) several comments are in order.

1. In contrast to the continuous case, the solutions of the Lippmann–Schwinger equation (2.16) depend on the reference material. This is, in part, caused by Γ_N^0 failing to be a projector, i.e. $\Gamma_N^0 : \Gamma_N^0 = \Gamma_N^0$ does not hold for clusters of finite size. In general, Γ_N^0 is only non-expansive in the sense that

$$\|\Gamma_N^0 : \mathbb{C}^0 : S\|_{\mathbb{C}^0} \leq \|S\|_{\mathbb{C}^0}, \quad S \in L^2(Y; \text{Sym}(d)) \quad (2.17)$$

holds. The non-expansivity (2.17) is a simple consequence of the non-expansivity of the projectors $\Gamma^0 : \mathbb{C}^0$ and Q_N , respectively, and the submultiplicativity of the operator norm.

2. If either $\alpha_+^0 < \alpha_-^0$ or $\alpha_-^0 > \alpha_+^0$ holds, the Hashin–Shtrikman variational principle (2.14) directly implies the existence and uniqueness to solutions of (2.16) for fixed $E \in \text{Sym}(d)$ and $\mathbb{C}^0$. However, it is known, see for instance Bornert’s thesis [51], that the best predictions of Hashin–Shtrikman type estimates are obtained if the reference material is chosen “close” to the material σ , in particular violating the inequalities mentioned in the beginning of this remark.
3. It is possible to conduct an analysis directly on the clustered Lippmann–Schwinger equation (2.16) to prove existence and uniqueness of solutions also for some reference materials which cannot be handled using the Hashin–Shtrikman variational principle. These reference materials need to commute with all local material tangents (which restricts the symmetry class of the reference material). Furthermore, the reference materials need to be sufficiently stiff for the analysis to be carried out, see also the discussion in Brisard–Dormieux [46]. To sum up, the approach just sketched is less general than the approach that we will pursue in Section 3.

3. Analysis of the clustered Lippmann–Schwinger equation

3.1. Existence and uniqueness

We wish to study the problem of existence and uniqueness for the clustered Lippmann–Schwinger equation (2.16), which we consider as given independently of any underlying variational principle. The conventional proof for the existence and uniqueness of Lippmann–Schwinger type equations, cf. Schneider [47], is inextricably tied to the projector property of the operator $\Gamma^0 : \mathbb{C}^0$. Thus, we proceed differently.

Eyre–Milton [52] introduced yet another equivalent reformulation of the corrector problem. For fixed stress operator σ and reference material $\mathbb{C}^0$, the Eyre–Milton identity holds

$$2\mathbb{C}^0 : [\text{Id} + \Gamma^0 : (\sigma - \mathbb{C}^0)](\sigma + \mathbb{C}^0)^{-1} = \text{Id} - Y^0 Z^0, \quad (3.1)$$

where $Y^0 = \text{Id} - 2\mathbb{C}^0 : \Gamma^0 \in L(L^2(Y; \text{Sym}(d)))$ and

$$Z^0 = (\sigma - \mathbb{C}^0)(\sigma + \mathbb{C}^0)^{-1}$$

defines a nonlinear operator on $L^2(Y; \text{Sym}(d))$. A derivation can be found in Schneider–Wicht–Böhlke [53].

In contrast to the Hashin–Shtrikman equation, for the Eyre–Milton identity (and the operator Z^0) σ and $\mathbb{C}^0$ are added (and not subtracted). Thus, for every reference material $\sigma + \mathbb{C}^0$ is a strongly monotone and Lipschitz continuous operator, which can be inverted (to a strongly monotone and Lipschitz continuous operator).

For fixed $E \in \text{Sym}(d)$ and reference material $\mathbb{C}^0$, the Eyre–Milton identity (3.1) directly implies that ε solves the Lippmann–Schwinger equation (2.7) if and only if $P = \sigma(\varepsilon) + \mathbb{C}^0 : \varepsilon$ solves the Eyre–Milton equation

$$(\text{Id} - Y^0 Z^0)P = 2\mathbb{C}^0 : E. \quad (3.2)$$

The structures of the Eyre–Milton equation and the Lippmann–Schwinger are formally similar. Both involve a non-local linear operator (Γ^0 and Y^0), a local non-linear operator ($\sigma - \mathbb{C}^0$ and Z^0) and a homogeneous right hand side (E and $2\mathbb{C}^0 : E$). The analysis of the Eyre–Milton equation (3.2) rests upon two pillars:

1. Y^0 is a reflection operator on $L^2(Y; \text{Sym}(d))$ and orthogonal w.r.t. $\langle \cdot, \cdot \rangle_{\mathbb{D}^0}$, where $\mathbb{D}^0 = (\mathbb{C}^0)^{-1}$. Both are proved by writing

$$\begin{aligned} Y^0 &= \text{Id} - 2\mathbb{C}^0 : \Gamma^0 \\ &= \mathbb{C}^0 : [\mathbb{D}^0 - 2\Gamma^0] \\ &= \mathbb{C}^0 : [\mathbb{D}^0 - 2\Gamma^0 : \mathbb{C}^0 : \mathbb{D}^0] \\ &= \mathbb{C}^0 : [\text{Id} - 2\Gamma^0 : \mathbb{C}^0] : \mathbb{D}^0 \\ &= \mathbb{C}^0 : ([\text{Id} - \Gamma^0 : \mathbb{C}^0] - \Gamma^0 : \mathbb{C}^0) : \mathbb{D}^0. \end{aligned}$$

Indeed, $\text{Id} - \Gamma^0 : \mathbb{C}^0$ and $\Gamma^0 : \mathbb{C}^0$ are complementary $\langle \cdot, \cdot \rangle_{\mathbb{C}^0}$ -orthogonal projectors. For our analysis, it is sufficient to note that Y^0 is non-expansive, i.e.

$$\|Y^0 : P\|_{\mathbb{D}^0} \leq \|P\|_{\mathbb{D}^0}, \quad P \in L^2(Y; \text{Sym}(d)), \quad (3.3)$$

holds.

2. Under the hypotheses (2.1) and (2.2), for every $\mathbb{C}^0$, Z^0 is a contraction operator on $(L^2(Y; \text{Sym}(d)), \langle \cdot, \cdot \rangle_{\mathbb{D}^0})$, i.e. Z^0 is Lipschitz continuous with Lipschitz constant ρ^0 strictly smaller than unity

$$\|Z^0 S - Z^0 T\|_{\mathbb{D}^0} \leq \rho^0 \|S - T\|_{\mathbb{D}^0}, \quad S, T \in L^2(Y; \text{Sym}(d)). \quad (3.4)$$

ρ^0 depends solely on the moduli of σ and the reference material $\mathbb{C}^0$. For better readability, a proof is out-sourced to Appendix.

With these two observations in hand, proving existence and uniqueness of solutions to the Eyre–Milton equation (3.2) is straightforward: the right hand side of the fixed point equation

$$P = 2\mathbb{C}^0 : E + Y^0 Z^0 P$$

defines a contraction on $(L^2(Y; \text{Sym}(d)), \langle \cdot, \cdot \rangle_{\mathbb{D}^0})$ with Lipschitz constant ρ^0 . The claim is implied by Banach's fixed point theorem [54].

The same techniques enable us to treat the clustered Lippmann–Schwinger equation (2.16)

$$\varepsilon_N + \Gamma_N^0 : (\sigma(\varepsilon_N) - \mathbb{C}^0 : \varepsilon_N) = E. \quad (3.5)$$

For $\varepsilon_N \in V_N$, the latter equation can be rewritten using the definition $\Gamma_N^0 = Q_N \Gamma^0 Q_N$ of Γ_N^0 and the compatibility (2.10) condition in the form

$$Q_N : [\text{Id} + \Gamma^0 : (\sigma - \mathbb{C}^0)] \varepsilon_N = E.$$

Applying the Eyre–Milton identity yields

$$Q_N : [\text{Id} - Y^0 Z^0] P_N = 2\mathbb{C}^0 : E$$

with $P_N = (\sigma + \mathbb{C}^0) \varepsilon_N$. As Z^0 commutes with Q_N , cf. (2.13), the latter equation can also be rewritten in the form

$$[\text{Id} - Y_N^0 Z^0] P_N = 2\mathbb{C}^0 : E \quad (3.6)$$

for $Y_N^0 = Q_N Y^0 Q_N$, which is very similar in structure to the (un-clustered) Eyre–Milton equation (3.2). As $\sigma + \mathbb{C}^0$ is invertible and preserves V_N , any solution ε_N of (3.5) automatically gives rise to a solution $P_N = \sigma(\varepsilon_N) + \mathbb{C}^0 : \varepsilon_N$ to (3.6) and vice versa. In particular, to show existence and uniqueness of solutions to (3.5) it suffices to establish existence and uniqueness of solutions to (3.6).

However, the question of existence and uniqueness of solutions to (3.6) is straightforward: rewriting (3.6) as a fixed point equation

$$P_N = Y_N^0 Z^0 P_N + 2\mathbb{C}^0 : E$$

the right hand side

$$F_N : V_N \rightarrow V_N, \quad S \mapsto Y_N^0 Z^0 S + 2\mathbb{C}^0 : E$$

is a contraction with contraction factor $\rho_0 < 1$ (as in (3.4)). Indeed, for any $S, T \in V_N$, we have

$$\begin{aligned} \|F_N(S) - F_N(T)\|_{\mathbb{D}^0} &= \|Y_N^0 Z^0 S - Y_N^0 Z^0 T\|_{\mathbb{D}^0} \\ &\leq \|Y_N^0\| \|Z^0 S - Z^0 T\|_{\mathbb{D}^0} \\ &\leq \rho_0 \|S - T\|_{\mathbb{D}^0}. \end{aligned} \quad (3.7)$$

In the last line, we used (3.4) and, by submultiplicativity of the operator norm,

$$\|Y_N^0\| = \|Q_N Y^0 Q_N\| \leq \|Q_N\|^2 \|Y^0\| = 1,$$

because both Q_N and Y^0 are orthogonal (and, thus, non-expansive).

By standard techniques of fixed point theory [55] the continuous dependence of the solution ε_N of the clustered Lippmann–Schwinger equation (2.16) both on the macroscopic strain E and the reference material $\mathbb{C}^0$ could be proved within our established framework. However, we will not pursue this here.

3.2. Convergence upon cluster refinement

In the previous section, the cluster decomposition $\{\Omega_i\}_{i=1}^N$ was fixed. In this section, we consider a sequence of cluster decompositions $\{\{\Omega_i^N\}_{i=1}^N\}$, parametrized by the number of clusters, N , with associated orthogonal projection operators $\{Q_N\}$. We call such a sequence of cluster decompositions convergent if, for every $S \in L^2(Y; \text{Sym}(d))$

$$\|Q_N S - S\|_{\mathbb{C}^0} \rightarrow 0 \quad (N \rightarrow \infty) \quad (3.8)$$

holds for some (and thus all) reference material(s) $\mathbb{C}^0$.

For fixed reference material $\mathbb{C}^0$, denote by ε_N^* and ε^* the unique solution of the clustered and the non-clustered Lippmann–Schwinger equation (2.16) and (2.7), respectively. Then, by continuity of $\sigma + \mathbb{C}^0$ and $(\sigma + \mathbb{C}^0)^{-1}$,

$$\|\varepsilon_N^* - \varepsilon^*\|_{\mathbb{C}^0} \rightarrow 0 \quad \text{as } N \rightarrow \infty \quad \text{if and only if} \quad \|P_N^* - P^*\|_{\mathbb{D}^0} \rightarrow 0 \quad \text{as } N \rightarrow \infty,$$

where P_N^* and P^* denote the unique solution to the clustered and non-clustered Eyre–Milton equations (3.6) and (3.2), respectively.

Subtracting (3.6) and (3.2) yields

$$\begin{aligned} P^* - P_N^* &= Y^0 Z^0 P^* - Y_N^0 Z^0 P_N^* \\ &= (\text{Id} - Q_N) Y^0 Z^0 P^* + Q_N Y^0 Z^0 (P^* - P_N^*). \end{aligned}$$

By the triangle inequality, by virtue of (3.7) and using the Eyre–Milton equation (3.2), we obtain

$$(1 - \rho_0) \|P^* - P_N^*\|_{\mathbb{D}^0} \leq \|(\text{Id} - Q_N) P^*\|_{\mathbb{D}^0}. \quad (3.9)$$

As $\rho_0 < 1$, independently of N , for any convergent cluster approximation, cf. (3.8), we obtain $P_N^* \rightarrow P^*$ in $L^2(Y; \text{Sym}(d))$ as $N \rightarrow \infty$. Thus, we have established convergence upon cluster refinement, as desired.

As a direct consequence, the convergence of the associated effective stresses

$$\langle \sigma(\varepsilon_N^*) \rangle_Y \rightarrow \langle \sigma(\varepsilon^*) \rangle_Y \quad \text{as } N \rightarrow \infty$$

can be proved, for instance by noting

$$\langle \sigma(\varepsilon_N^*) \rangle_Y - \langle \sigma(\varepsilon^*) \rangle_Y = \langle P_N^* - P^* \rangle_Y,$$

and applying the Cauchy–Schwarz inequality.

Let us close this section by remarking that the estimate on the convergence rate (3.9) is sharp. With little effort, it can be strengthened to

$$(1 - \rho_0) \|P^* - P_N^*\|_{\mathbb{D}^0} \leq \|(\text{Id} - Q_N) P^*\|_{\mathbb{D}^0} \leq (1 + \rho_0) \|P^* - P_N^*\|_{\mathbb{D}^0}.$$

Thus, for fixed reference material, the convergence rates of $\|P^* - P_N^*\|_{\mathbb{D}^0}$ and $\|(\text{Id} - Q_N) P^*\|_{\mathbb{D}^0}$ are identical.

4. Relation to Dvorak’s transformation field analysis for small strain elasto-viscoplasticity

In this mathematically non-rigorous section we take a closer look at the clustered Lippmann–Schwinger equation for the special case of elasto-viscoplastic material at small strains, and establish a connection of the transformation field analysis, originally introduced by Dvorak–Benveniste [25].

Let us fix a cluster decomposition $\{\mathcal{Q}_i\}$ and assume a small strain elasto-viscoplastic generalized standard material [56] is given, which furthermore respects the cluster decomposition. Fix a dimension n for the hardening variable q , and suppose elastic tensors $\mathbb{C}_i$, kinematic hardening potentials $g_i : \text{Sym}(d) \rightarrow \mathbb{R}$ as well as isotropic hardening potentials $h_i : \mathbb{R}^n \rightarrow \mathbb{R}$ and dissipation potentials $\phi_i : \text{Sym}(d) \times \mathbb{R}^n \rightarrow \mathbb{R}$ are given. Suppose, furthermore, that a time discretization $t_0 < t_1 < \dots$ is fixed. Then, for prescribed initial values $\varepsilon_0^p : Y \rightarrow \text{Sym}(d)$ and $q_0 : Y \rightarrow \mathbb{R}^n$ the time-discretized evolution of the multi-scale generalized standard elastoviscoplastic material is governed by the balance of linear momentum

$$\text{div } \mathbb{C} : (\varepsilon_n - \varepsilon_n^p) = 0, \quad n = 0, 1, 2, \dots$$

with the total strain $\varepsilon_n = E_n + \nabla^s u_n$ involving a displacement fluctuation field $u_n \in H_{\#}^1(Y; \mathbb{R}^d)$ and Biot’s equation

$$(\mathbb{C} : (\varepsilon_{n+1} - \varepsilon_{n+1}^p) - g'(\varepsilon_{n+1}^p), -h'(q_{n+1})) \in \partial \phi \left(\frac{\varepsilon_{n+1}^p - \varepsilon_n^p}{t_{n+1} - t_n}, \frac{q_{n+1} - q_n}{t_{n+1} - t_n} \right).$$

Here, $E_n \in \text{Sym}(d)$ denotes the applied macroscopic strain at time t_n and we have introduced

$$\mathbb{C} = \sum_{i=1}^N \mathbb{C}_i \chi_i, \quad g = \sum_{i=1}^N g_i \chi_i, \quad h = \sum_{i=1}^N h_i \chi_i, \quad \text{and} \quad \phi = \sum_{i=1}^N \phi_i \chi_i.$$

For fixed time step, indexed by $n \geq 0$, Biot’s equation and the balance of linear momentum can be obtained from a variational principle, see Ortiz–Stainier [57] or Miehe [58]. More precisely, $(u_{n+1}, \varepsilon_{n+1}^p, q_{n+1})$ are critical points of the functional

$$\begin{aligned} J_{n+1}(u, \varepsilon^p, q) &= \int_Y \frac{1}{2} (E_n + \nabla^s u - \varepsilon^p) : \mathbb{C} : (E_n + \nabla^s u - \varepsilon^p) + g(\varepsilon^p) + h(q) \\ &\quad + \Delta t_n \phi \left(\frac{\varepsilon^p - \varepsilon_n^p}{\Delta t_n}, \frac{q - q_n}{\Delta t_n} \right) dx, \end{aligned} \quad (4.1)$$

where $\Delta t_n = t_{n+1} - t_n$. For discretization by the transformation field analysis, we proceed, see Michel–Suquet [59], by restricting to cluster-wise homogeneous internal variables (ε^p, q) , i.e. we consider a Galerkin discretization of (4.1) w.r.t.

$$\varepsilon_n^p = \sum_{i=1}^N \varepsilon_{n,i}^p \chi_i \quad \text{and} \quad q_n = \sum_{i=1}^N q_{n,i} \chi_i,$$

and $u_n \in H_{\#}^1(Y; \mathbb{R}^d)$ is left unrestricted. For this discretization, the Euler–Lagrange equations read

$$\operatorname{div} \mathbb{C} : (\varepsilon_{n+1} - \varepsilon_n^p) = 0 \quad (4.2)$$

and

$$(\mathbb{C}_i : (\langle \varepsilon_{n+1} \rangle_{\Omega_i} - \varepsilon_{n+1,i}^p) - g'_i(\varepsilon_{n+1,i}^p), -h'_i(q_{n+1,i})) \in \partial \phi_i \left(\frac{\varepsilon_{n+1,i}^p - \varepsilon_{n,i}^p}{\Delta t_n}, \frac{q_{n+1,i} - q_{n,i}}{\Delta t_n} \right) \quad (4.3)$$

for all $i = 1, \dots, N$, where $\langle \cdot \rangle_{\Omega_i}$ denotes averaging on Ω_i . Thus, the TFA discretization proceeds by assuming cluster-wise homogeneous internal variables, which are driven by the phase-wise averaged strains which are obtained by solving the eigenstrain problem (4.2). To be closer to the self-consistent clustering analysis, for any reference material $\mathbb{C}^0$ the balance of linear momentum (4.2) can be equivalently written as a Lippmann–Schwinger equation

$$\varepsilon_{n+1} + \Gamma^0 : (\mathbb{C} : (\varepsilon_{n+1} - \varepsilon_n^p) - \mathbb{C}^0 : \varepsilon_{n+1}) = E_{n+1}.$$

By comparison, for given $\mathbb{C}^0$, the self-consistent clustering analysis seeks a sequence $(\varepsilon_n, \varepsilon_n^p, q_n)$, which inductively solves

$$\varepsilon_{n+1} + \Gamma_N^0 : (\mathbb{C} : (\varepsilon_{n+1} - \varepsilon_n^p) - \mathbb{C}^0 : \varepsilon_{n+1}) = E_{n+1}$$

and (4.3) for all $i = 1, \dots, N$. In this formulation, we see that the only difference between the self-consistent clustering analysis and the transformation field analysis for small strain elasto-viscoplastic generalized standard materials is whether the clustered or the non-clustered Eshelby–Green operator is used.

5. Numerical investigations

5.1. Setup and materials

The self-consistent clustering analysis (SCA) was implemented as a post-processor to an in-house FFT-based computational micromechanics code [60,61]. For the simulations at hand, we relied upon the staggered grid discretization [62].

Six linear elastic computations were carried out to identify the strain localization tensor, as suggested in Liu–Bessa–Liu [36]. Subsequently, the clusters were identified by Lloyd’s k -means algorithm [44], accelerated by using the triangle inequality as suggested by Elkan [63], based on the strain localization tensor. The initial clusters were chosen randomly based on the voxel index. In contrast to what was reported in Liu–Bessa–Liu [36], repeated restarting had negligible effect on the approximation quality. The relative Frobenian error of the strain localization tensor was used to measure the quality of the clustering, and different randomly chosen initial positions only changed the quality measure on the third significant digit.

The components of the discretized Eshelby–Green operator (2.15) Γ_N w.r.t. the basis $\{\chi_i e_\alpha\}$, where $\{\chi_i\}_{i=1}^N$ stands for the characteristic functions of the clusters and $\{e_\alpha\}_{\alpha=1}^6$ denote a basis of $\operatorname{Sym}(3)$ used for Voigt’s notation [64], were also computed by FFT-based methods and stored in a database. Notice that we only use reference materials proportional to the identity Id of $\operatorname{Sym}(3)$.

This database was used in solving the clustered Lippmann–Schwinger equation by Newton’s method, where the reference material was chosen to be closest to the effective material tangent, as suggested in Liu–Bessa–Liu [36].

The inelastic computations used for comparison were also conducted by FFT-based micromechanics. For the numerical investigation, a polyamide (PA) with glass fillers was used. The E-glass is modeled as linear elastic, and the mechanical behavior of the PA-matrix is governed by J_2 -plasticity with exponential-linear hardening

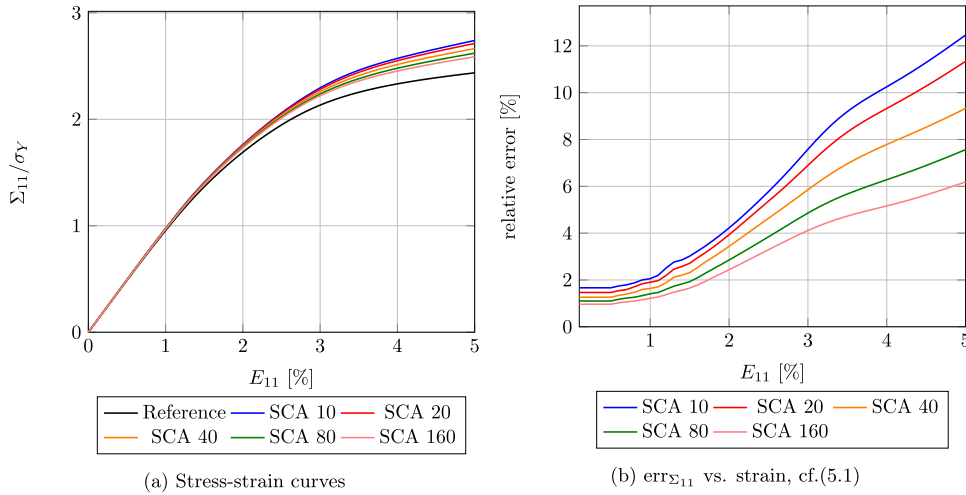
$$\sigma_0(p) = \sigma_Y + k_1 p + k_2(1 - \exp(-mp)).$$

The material parameters used in the simulations are summarized in Table 1.

Table 1

Material parameters for the short glass fiber reinforced polyamide, cf. Doghri et al. [65].

E-glass fillers	$E = 72$ GPa,	$\nu = 0.22$				
PA matrix	$E = 2.1$ GPa,	$\nu = 0.3$,	$\sigma_Y = 29$ MPa,	$k_1 = 139$ MPa,	$k_2 = 32.7$ MPa,	$m = 319.4$

**Fig. 2.** Performance of SCA on single glass inclusion in a PA matrix. Absolute and relative axial stress vs. axial strain.

5.2. A spherical glass inclusion

We investigate a spherical E-glass inclusion in a PA matrix with a filler volume fraction of 12.9%. The microstructure is discretized on a 32^3 grid.

We examine uniaxial extension in 1-direction up to 5% axial strain, i.e. $E_{11} = 5\%$, and the stresses orthogonal to $e_x \otimes e_x$ are forced to zero, distributed over 50 equidistant load steps.

Liu–Bessa–Liu [36] suggested that the ratio of the clusters for different phases should roughly equal the volume fraction. Thus, as the smallest setup we chose 9 : 1 clusters for the matrix and the E-glass phase, respectively. Then, we increased the number of clusters iteratively by 2 keeping the ratio 9 : 1.

The effective axial stress (normalized by the microscopic yield stress) vs. the effective axial strain is shown in Fig. 2(a) for different cluster counts and in comparison to the reference full field simulation. We observe that the initial elastic phase (up to $\approx 1.5\%$ strain) is captured well, and the SCA overestimates the stress thereafter. Furthermore, increasing the number of clusters consistently improves the stress prediction.

To quantify these qualitative observations we take a closer look at the relative error in axial stress

$$\text{err}_{\Sigma_{11}} = \|\Sigma_{11} - \Sigma_{11}^{\text{SCA}}\| / \|\Sigma_{11}\| \quad (5.1)$$

as presented in Fig. 2(b). We see that the monotonicity behavior of the error curves for different cluster counts is similar. Up to 2% strain, all investigated cluster counts yield a relative error smaller than 5%. For higher strains, the relative error increases quickly. Only the SCA variant with 160 clusters manages to stay below 5% relative error up to 4% axial strain. To gain insight into the convergence behavior of the SCA discretization scheme we take a look at the relative stress error for various macroscopic strain levels and varying cluster counts, cf. Fig. 3(a). For all considered strain levels, the relative error exhibits a logarithmic rate of convergence, i.e. a linear curve in our plot where the E_{11} -axis is stretched logarithmically. For comparison, the relative error of approximating the strain localization tensor is shown. Apparently, the convergence rate for the latter is also logarithmic, but much better than the relative stress errors. We affiliate this phenomenon to the fact that it might be better to cluster based on the strain localization tensor after plastification occurred, and not based on the purely elastic strain localization tensor.

For the convenience of the reader, a plot for the local plastic strain at 5% on a cross-section was included in Fig. 3(b). Except for oscillations at the material interface (which are a well-known phenomenon in FFT-based micromechanics), the local plastic strain is rather uniform.

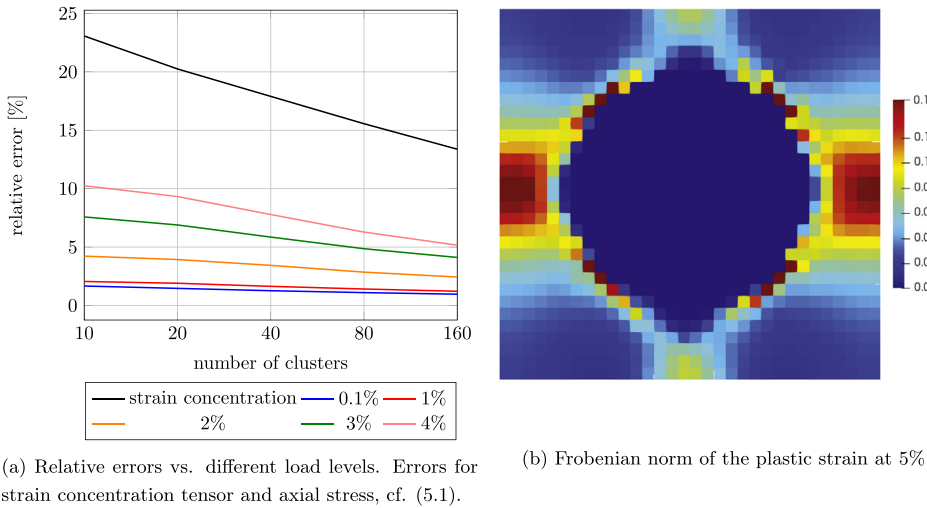


Fig. 3. Performance of SCA on single glass inclusion in a PA matrix. Convergence rate plots of the axial stress at different levels of applied axial strain in comparison to the relative approximation error of the strain localization tensor. Furthermore, the (norm of the) plastic strain at 5% loading is shown.

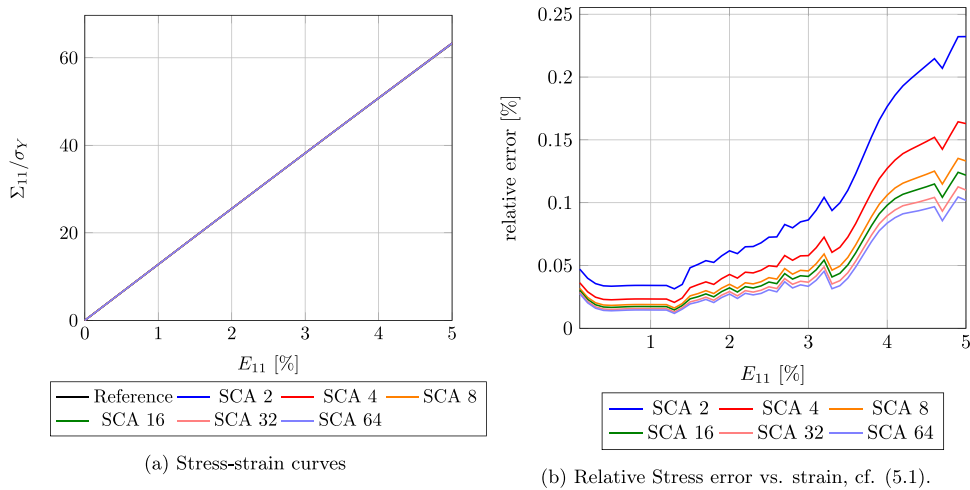


Fig. 4. Performance of SCA on a continuous glass fiber reinforced PP and uniaxial extension in fiber direction. Absolute and relative axial stress vs. axial strain.

5.3. A continuous fiber reinforced microstructure

We consider a continuously fiber reinforced microstructure with 50% E-glass fiber volume fraction, and discretized on an 512×512 pixel grid. The material properties of glass and PA are identical to the previous section. For the SCA, the number of clusters per phase is chosen to be identical, i.e. we use the ratio 1 : 1 for the number of clusters in the respective phase. We start with 1 + 1 clusters and increase them repeatedly by a factor of 2 until 32 + 32 clusters.

First, we investigate uniaxial extension in fiber direction up to 5% strain, discretized by 50 equidistant time steps. The stress-strain curve, cf. Fig. 4(a), reveals that macroscopically, the behavior is almost perfectly linear. More precisely, see also Fig. 5(b), the maximal norm of the plastic strain at the micro scale for 5% loading is at 4%, which is comparatively low. It is also apparent from Fig. 4(a) that the stress-strain curves for different number of clusters are indistinguishable from the reference computation. To quantify this observation, a closer look at the

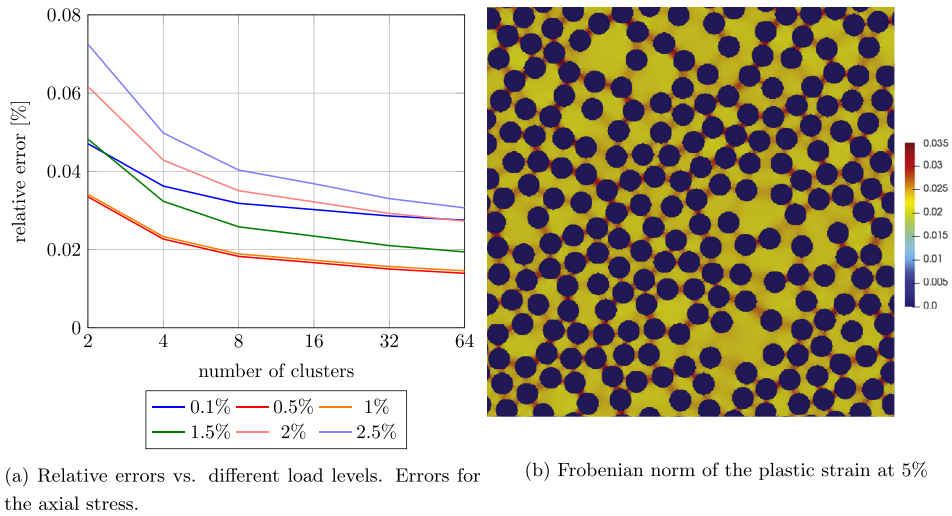


Fig. 5. Performance of SCA on a continuous glass fiber reinforced PP and uniaxial extension in fiber direction. Convergence rate plots of the axial stress at different levels of applied axial strain. Furthermore, the (norm of the) plastic strain at 5% loading is shown.

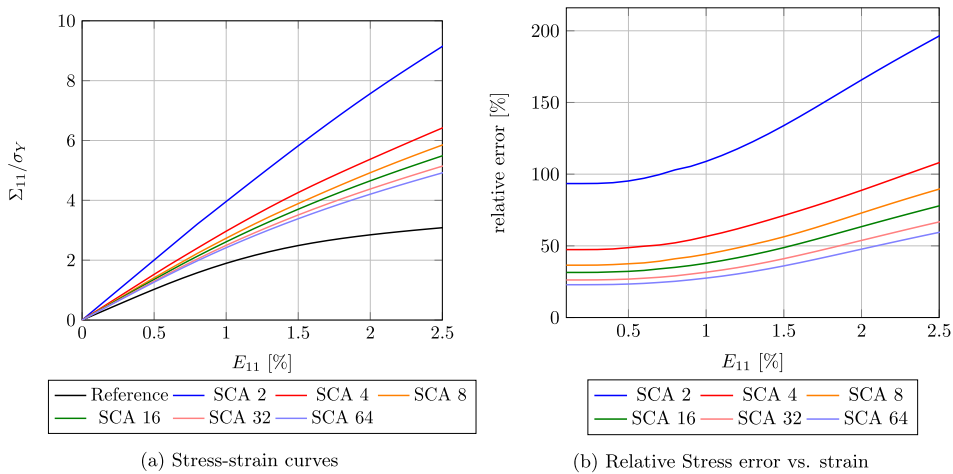


Fig. 6. Performance of SCA on a continuous glass fiber reinforced PP and uniaxial extension transverse to the fiber direction. Absolute and relative axial stress vs. axial strain.

relative errors in axial stress, cf. Fig. 4(b), reveals that already the 1 + 1 cluster version of the SCA does not exceed a relative error of 0.25%. Taking a closer look at the convergence behavior of the relative stress error for different levels of applied strain, cf. Fig. 5(a), reveals, again, a logarithmic and monotonic convergence behavior (for at least 8 clusters).

Next, we investigate uniaxial loading transverse to the fiber direction, up to 2.5% strain, equidistantly distributed in 25 load steps. The stress–strain curve, cf. Fig. 6(a), reveals a strong plastification. The SCA results for different number of clusters overestimate the stress strongly. By increasing the number of clusters, the overall error decreases. These observations are quantified in Fig. 6(b). Already in the elastic regime, the errors are large. Assuming piecewise constant strains (1 + 1 clusters) overpredicts the stresses almost by a factor of 2. For the largest number of considered clusters, the relative stress error is 22.89%.

For all considered cluster counts, the relative error increases with increasing strain, owing to the increased heterogeneity of the strain and stress fields. To illustrate this heterogeneity, the plastic strain at 2.5% strain is shown in Fig. 7(b). For better visibility, the values are clipped to 10%. The maximal $\|\varepsilon^P\|$ for this level of loading is 44%. From the viewpoint of continuum mechanics, this level is way beyond our small strain assumption. Still, in

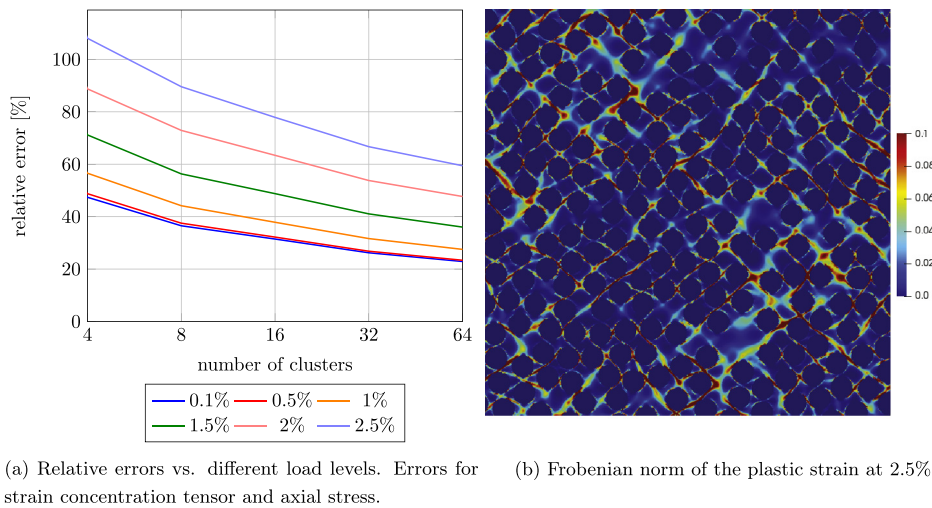


Fig. 7. Performance of SCA on a continuous glass fiber reinforced PP and uniaxial extension transverse to the fiber direction. Convergence rate plots of the axial stress at different levels of applied axial strain. Furthermore, the (norm of the) plastic strain at 5% loading is shown.

computational micromechanics, it is not uncommon to use small-strain models also if finite strains would be more appropriate. The reason is that, in many cases, finite strain effects on the microscale do not reflect in the macroscopic material response, see Schüller et al. [66]. Notice that already at 0.7% applied strain, locally a maximum of 5.38% plastic strain (in norm) is reached. The convergence of the relative error in axial stress is shown in Fig. 7(a). The error decreases monotonically with increasing cluster count, and the convergence rate is linear in the log-plot. Also, there is almost no difference in relative error between a load level of 0.1% strain and 0.5% strain, although the local plastic strain field is identically zero for the 0.1% applied macroscopic strain. Thus, we conclude that the large errors for the transverse loading already originate from the linear elastic behavior, which is insufficiently captured by the clustering.

6. Conclusion

This work analyzed the recently introduced self-consistent clustering analysis (SCA) from a mathematical point of view. For arbitrary reference material and material laws whose time discretized stress–strain relationships satisfy the inequalities (2.1) and (2.2), the clustered Lippmann–Schwinger equation admits a unique solution. In particular, the model is well-defined and dedicated numerical techniques, like Newton’s method, can be applied.

Our technique of derivation was inspired by recent progress in understanding polarization schemes in FFT-based computational micromechanics [53], rewriting the clustered Lippmann–Schwinger equation as a clustered Eyre–Milton equation. Existence and uniqueness of the Eyre–Milton equation are easily established for reference materials of arbitrary anisotropy and strongly monotone and Lipschitz continuous stress operators. Of course, the associated Eyre–Milton iteration [52] gives rise to an efficient numerical solution technique — a research direction which we did not pursue in this manuscript. Furthermore, incorporating reference materials more general than isotropic [36,43] could improve the SCA.

The material laws for which our result is valid include all small strain materials for which the material tangent is symmetric and whose eigenvalues are bounded below by a constant independent of the applied strain. Thus, we cover a large portion of strain-hardening materials of interest. It would be interesting to extend our result to the case of Lipschitz-continuous but merely strictly convex stress operators. In the continuous setting, existence and uniqueness of solutions to the corrector problem (2.5) are immediate from Minty–Browder’s Theorem [67,68]. However, the approach based on the Eyre–Milton equation appears to be limited to the strongly convex setting, as in Giselsson–Boyd [69]. From a physical point of view, one would expect existence and uniqueness of solutions to the clustered Lippmann–Schwinger equation to hold. However, this might require methods which go beyond those of this manuscript.

In addition to the existence and uniqueness result, this work also includes a proof of the convergence of the solution to the clustered Lippmann–Schwinger equation to the solution of the (continuous) Lippmann–Schwinger equation upon cluster refinement. The convergence estimate (3.9) is explicit and involves the L^2 approximation of the polarization field $P = \sigma + \mathbb{C}^0 : \varepsilon$. We did not see how (3.9) could give us a direct hint for choosing the reference material, but this issue could be subject to further investigation. However, the estimate (3.9) motivates studying the literature on the approximation of general functions by piece-wise constant functions. Mathematicians are familiar with this type of setting — essentially because Lebesgue’s integral is defined by piece-wise constant approximation of Lebesgue-measurable functions. A recent survey article by Davydov [70] states that, even for functions of high smoothness, only logarithmic convergence can be expected. Also, the complexity of interpolation by piecewise constant functions is the same as for approximation on voxel grids (with different prefactors, of course). These theoretical predictions were confirmed by our numerical examples, although the required assumptions on the regularity of the functions to be approximated are not (known to be) satisfied.

We exhibit examples where the self-consistent clustering examples did not perform exceedingly well. Still, we believe that there is an area of application of the SCA, in particular for materials with low elastic contrasts or low filler fraction. Of particular importance is the generality of the SCA — there is no technological restriction of the class of material behavior which can be treated by the SCA. This contrasts strongly with the transformation field analysis of Dvorak–Benveniste [25], which is similar in spirit (and, possibly, the formulation), but is restricted to small strain elasto-viscoplasticity.

The mathematical framework established in this article is expected to advance the further development of the self-consistent clustering analysis, as observed for other mathematical justifications of insights from engineering, like finite element analysis or FFT-based homogenization methods.

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Appendix. Contractivity of Z^0

In this section, we wish to show that, for fixed reference material $\mathbb{C}^0$ and stress operator σ satisfying (2.1) and (2.2), the operator $Z^0 = (\sigma - \mathbb{C}^0)(\sigma + \mathbb{C}^0)^{-1}$ defines a contraction on $L^2(Y; \text{Sym}(d))$, equipped with the inner product $\langle \cdot, \cdot \rangle_{\mathbb{D}^0}$.

For that purpose, we make use of recent work by Giselsson–Boyd [69]. They proved that for any convex function g on a Hilbert space H , which is c -strongly monotone and has C -Lipschitz continuous (Frechét-) gradient, and $\lambda > 0$, the operator

$$(\nabla f - \lambda \text{Id})(\nabla f + \lambda \text{Id})^{-1}$$

is contractive on H with contractivity constant $\gamma = \max\left(\frac{C/\lambda-1}{C/\lambda+1}, \frac{1-c/\lambda}{1+c/\lambda}\right)$, cf. Proposition 2 in Giselsson–Boyd [69].

For the remainder of this section, we fix $\mathbb{C}^0$. As $\mathbb{C}^0$ is symmetric and positive, the operator square root can be computed to give

$$\mathbb{C}^0 = \mathbb{B} : \mathbb{B}.$$

(As $\mathbb{C}^0$ remains fixed throughout this argument, we suppress the superscript 0 at $\mathbb{B}$ for better readability.) With the help of $\mathbb{B}$, the $\mathbb{C}^0$ - and $\mathbb{D}^0 = (\mathbb{C}^0)^{-1}$ -weighted L^2 norms can be expressed in terms of the unweighted L^2 -norm

$$\|S\|_{\mathbb{C}^0} = \|\mathbb{B} : S\|_{\text{Id}} \quad \text{and} \quad \|Y\|_{\mathbb{D}^0} = \|\mathbb{B}^{-1} : T\|_{\text{Id}} \quad \text{for all } S, T \in L^2(Y; \text{Sym}(d)). \quad (6.1)$$

Writing

$$\sigma - \mathbb{C}^0 = \sigma - \mathbb{B}^2 = \mathbb{B} : (\tilde{\sigma} - \text{Id}) : \mathbb{B} \quad \text{with} \quad \tilde{\sigma} = \mathbb{B}^{-1} \sigma \mathbb{B}^{-1}$$

implies

$$Z^0 = (\sigma - \mathbb{C}^0)(\sigma + \mathbb{C}^0)^{-1} = \mathbb{B}(\tilde{\sigma} - \text{Id})(\tilde{\sigma} + \text{Id})^{-1}\mathbb{B}^{-1}.$$

Thus, we have, for any $S, T \in L^2(Y; \text{Sym}(d))$,

$$\begin{aligned} \|Z^0 S - Z^0 T\|_{\mathbb{D}^0} &= \|\mathbb{B}^{-1} : (Z^0 S - Z^0 T)\|_{\text{Id}} \\ &= \|(\tilde{\sigma} - \text{Id})(\tilde{\sigma} + \text{Id})^{-1}\mathbb{B}^{-1}S - (\tilde{\sigma} - \text{Id})(\tilde{\sigma} + \text{Id})^{-1}\mathbb{B}^{-1}T\|_{\text{Id}} \\ &\leq \text{Lip}((\tilde{\sigma} - \text{Id})(\tilde{\sigma} + \text{Id})^{-1})\|\mathbb{B}^{-1}S - \mathbb{B}^{-1}T\|_{\text{Id}} \\ &\leq \text{Lip}((\tilde{\sigma} - \text{Id})(\tilde{\sigma} + \text{Id})^{-1})\|S - T\|_{\mathbb{D}^0}, \end{aligned}$$

where $\text{Lip}(A)$ denotes the Lipschitz constant of a (nonlinear) operator A on $L^2(Y; \text{Sym}(d))$ w.r.t. $\|\cdot\|_{\text{Id}}$. As σ arises as the (Frechét-)derivative of a convex function ψ on $L^2(Y; \text{Sym}(d))$, cf. the beginning of Section 2.1, $\tilde{\sigma}$ arises as the (Frechét-)derivative of the function

$$\tilde{\psi} : L^2(Y; \text{Sym}(d)) \rightarrow \mathbb{R}, \quad \varepsilon \mapsto \psi(\mathbb{B}^{-1}\varepsilon),$$

which is convex, as ψ is convex and $\mathbb{B}^{-1}$ is symmetric and positive definite. Also, $\tilde{\sigma}$ inherits strong convexity and Lipschitz continuity from σ (but with different constants). Thus, the claim follows by Proposition 2 in Giselsson–Boyd [69].

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